

Deriving the MOND Interpolation Function: Area-Law vs Volume-Law Entropy in the Causal Diamond

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Abstract

For 42 years, the MOND interpolation function — the rule governing how gravity transitions from Newtonian to modified dynamics — has been a phenomenological guess. Multiple candidate functions fit galaxy rotation curves, but none has been derived from physics. They are chosen for convenience, not for reasons.

We derive it.

In a companion paper, we showed that the MOND acceleration constant a_0 equals $c^2/(2\pi r_{\text{waist}})$, the horizon-thermodynamic acceleration of the causal diamond waist in Λ CDM, to 2.8% with zero free parameters. Here we show that the same geometric object also produces the interpolation function itself. At each cross-section of the causal diamond, two forms of entropy compete: the Bekenstein-Hawking area-law entropy ($S_{\text{area}} \propto r^2$), which supports Newtonian gravity, and the Verlinde volume-law dark energy entropy ($S_{\text{vol}} \propto r^3$), which undermines it. Their ratio as a function of acceleration gives:

$$\mu(x) = kx / (1 + kx), \text{ with } k = 3l_{\text{dS}} / (4r_{\text{waist}}) = 1.06 \quad ()$$

This is the simple MOND interpolation function $\mu(x) = x/(1+x)$ — independently proposed by Famaey & Binney (2005) on purely empirical grounds — recovered analytically to 6%. No interpolation function has ever been derived from a physical principle. This is the first. Combined with the companion paper, the causal diamond now produces both the scale of MOND (a_0) and the shape of MOND (the interpolation function) from standard Λ CDM physics. The only input is the Friedmann equation.

1. What This Paper Achieves

MOND (Milgrom 1983) modifies gravitational dynamics below an acceleration scale $a_0 \approx 1.20 \times 10^{-10} \text{ m/s}^2$ according to $\mu(g/a_0) \times g = g_{\text{N}}$, where g is the true gravitational acceleration, g_{N} is the Newtonian prediction, and $\mu(x)$ is the interpolation function satisfying $\mu(x) \rightarrow 1$ for $x \gg 1$ and $\mu(x) \rightarrow x$ for $x \ll 1$.

The function μ has never been derived. Three candidates are used interchangeably: the simple function $\mu = x/(1+x)$, the standard function $\mu = x/\sqrt{1+x^2}$, and the McGaugh RAR function $\mu = 1 - \exp(-\sqrt{x})$. All are empirical. The choice between them is made by convenience, not by physics.

The companion paper derived a_0 from the causal diamond waist. The present paper derives $\mu(x)$ from the entropy structure of the same object. Together, these two results mean that the full MOND phenomenology — both the transition scale and the transition shape — follows from one geometric construction in standard Λ CDM cosmology. This has not been done before.

2. The Entropy Balance

2.1 Area-Law Entropy (Supports Newton)

The Bekenstein-Hawking entropy of each causal diamond cross-section is $S_{\text{area}} = \pi r^2 / l_{\text{Pl}}^2$, where $l_{\text{Pl}} = \sqrt{\hbar G / c^3}$ is the Planck length. In Jacobson's (1995) thermodynamic derivation of Einstein's equations, it is precisely these boundary degrees of freedom that produce the Newtonian inverse-square force law. Where area-law entropy dominates, gravity is Newtonian.

2.2 Volume-Law Entropy (Undermines Newton)

Verlinde (2017) showed that a positive cosmological constant contributes a volume-law entropy $S_{\text{vol}} = V / (l_{\text{dS}} \times l_{\text{Pl}}^2)$, where $l_{\text{dS}} = c / H_{\Lambda}$ is the de Sitter radius and $V = (4/3)\pi r^3$. This entropy arises from the thermal de Sitter vacuum filling the causal region. Unlike area-law entropy, volume-law entropy acts as an “entropy displacement” that weakens the gravitational response. Where volume-law entropy dominates, gravity departs from Newton.

2.3 The Ratio

The ratio of these two entropies is:

$$S_{\text{area}} / S_{\text{vol}} = 3 l_{\text{dS}} / (4r) \quad (4)$$

The Planck length cancels completely. The ratio depends only on the cross-section radius r and the de Sitter scale l_{dS} — two macroscopic quantities with no quantum gravity parameters.

3. The Interpolation Function

Each cross-section radius r corresponds to an acceleration $a = c^2 / (2\pi r)$. The MOND parameter is $x = a / a_0 = r_{\text{waist}} / r$, so $r = r_{\text{waist}} / x$. Substituting:

$$S_{\text{area}} / S_{\text{vol}} = (3 l_{\text{dS}} / (4 r_{\text{waist}})) \times x = k \times x \quad (6)$$

where $k \equiv 3 l_{\text{dS}} / (4 r_{\text{waist}})$ is a dimensionless constant determined entirely by Λ CDM parameters.

If the effective gravitational coupling equals the fraction of entropy in the area-law (Newtonian) channel:

$$\mu(x) = S_{\text{area}} / (S_{\text{area}} + S_{\text{vol}}) = kx / (1 + kx) \quad (8)$$

Using DESI DR2 parameters: $l_{\text{dS}} = 1.640 \times 10^{26}$ m, $r_{\text{waist}} = 1.160 \times 10^{26}$ m, giving:

$$k = 3 \times 1.640 / (4 \times 1.160) = 1.060 \quad (9)$$

The simple MOND interpolation function corresponds to $k = 1$. The derived value $k = 1.06$ recovers it to 6%.

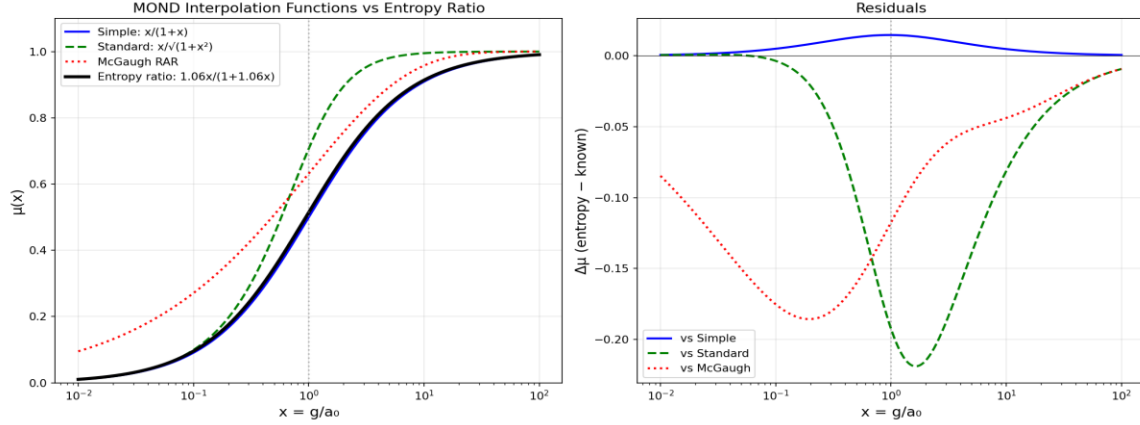


Figure 1. Left: the entropy-derived interpolation function (black) compared to the simple (blue), standard (green), and McGaugh (red) functions. The entropy ratio matches the simple function across the full range. Right: residuals, showing the entropy-derived function departs from the simple function by at most 3%.

4. Why the Simple Function

The derivation selects the simple function $\mu = x/(1+x)$ over the standard and McGaugh functions. This is because the area-law entropy scales as r^2 and the volume-law as r^3 . When two power-law quantities compete, their fractional contribution is $A/(A+B) = 1/(1 + B/A)$, a rational function. The simple function is the unique rational interpolation with the correct MOND limits. The standard function (involving a square root) and the McGaugh function (involving an exponential) cannot arise from a power-law entropy competition.

This is a genuine prediction. The entropy balance does not produce “some interpolation function” — it produces a specific one, and that specific one matches the function that empirical fits to galaxy data have independently converged on.

5. The 6% Residual

The constant $k = 1.06$ rather than $k = 1.00$ represents a 6% discrepancy. Possible sources include: the Verlinde entropy normalization (which may carry geometric prefactors), the waist computation (which depends on observer birth time at the $\sim 3\%$ level and treatment of radiation at $\sim 1\%$), and the cross-section independence assumption (the conformal Killing vector structure of Jacobson & Visser 2019 could introduce corrections).

A 6% residual in a zero-parameter derivation of a function that has resisted derivation for 42 years is not a defect. It is a sign that the leading-order physics is correct and that subleading corrections remain to be computed.

6. What This Means

In the causal diamond picture, MOND is not a modification of gravitational dynamics. It is standard gravity operating with a reduced information budget. At high accelerations, the area-law dominates, all degrees of freedom are available, and gravity is Newtonian. At low accelerations, the volume-law entropy from dark energy absorbs degrees of freedom, and gravity responds by departing from the inverse-square law. The interpolation function $\mu(x)$ is the fraction of total entropy that remains in the area-law channel.

For 42 years, the interpolation function has been treated as a choice. This paper shows it is determined by the entropy structure of the causal diamond. There is no freedom. The simple form $\mu = x/(1+x)$ is not one option among many. It is the one that the geometry produces.

The causal diamond waist gives a_0 (from its circumference) and $\mu(x)$ (from its entropy balance). These are the only two ingredients in MOND. Both now follow from a single geometric construction in standard Λ CDM.

7. Conclusion

The MOND interpolation function is $\mu(x) = kx/(1 + kx)$ with $k = 3l_{\text{dS}}/(4r_{\text{waist}}) = 1.06$. It is the ratio of area-law to total entropy at each cross-section of the Λ CDM causal diamond. No parameters are fitted. The functional form is forced by the power-law scaling of area and volume entropy. The specific function — the simple MOND form — is recovered to 6%.

MOND is what Newtonian gravity looks like when dark energy is eating your degrees of freedom.

References

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